

Cafe Sales Performance and Analytics

Mini Project Report — Prepared by Ansania Salam

Phase 1: Data Pre-processing using Microsoft Excel

- Imported the cafe sales transaction dataset (10,000 records) into Microsoft Excel and organized the data for preprocessing.
- Checked the dataset for duplicate records using the Transaction ID column.
- Inspected all columns for missing values, including blank cells and explicit “UNKNOWN” and “ERROR” placeholder text, using Excel filters.

Item

- Identified missing and invalid values in the Item column (blank, “UNKNOWN,” and “ERROR” entries).
- Replaced all such entries with “Unknown” using Find & Replace to standardize the column while keeping every transaction row.

Quantity

- Checked the Quantity column for missing, blank, “UNKNOWN,” and “ERROR” values using Excel filters.
- Calculated the median of the available quantity values using the MEDIAN() function.
- Used the calculated median to fill the missing entries while preserving all originally valid values.
- Verified that the Quantity column contained no missing values after preprocessing.

Price Per Unit

- Checked the Price Per Unit column for missing, blank, “UNKNOWN,” and “ERROR” values using Excel filters.
- Calculated the median of the available price values using the MEDIAN() function.
- Used the calculated median to fill the missing entries while keeping the existing values unchanged.
- Verified that no missing values remained in the Price Per Unit column.

Total Spent

- Checked the Total Spent column for missing, blank, “UNKNOWN,” and “ERROR” values.
- Rather than imputing Total Spent independently, recalculated it for every row using $\text{Quantity} \times \text{Price Per Unit}$, ensuring the column stayed fully consistent once Quantity and Price Per Unit had been cleaned.
 =C2*D2
- Verified that no missing values remained in the Total Spent column.

Payment Method

- Checked the Payment Method column for missing, blank, “UNKNOWN,” and “ERROR” values.
- Replaced all such entries with “Unknown” using Find & Replace.

Location

- Checked the Location column for missing, blank, “UNKNOWN,” and “ERROR” values.
- Replaced all such entries with “Unknown” using Find & Replace.

Transaction Date

- Checked the Transaction Date column for missing, blank, “UNKNOWN,” and “ERROR” values, and found a significant number of invalid entries.
- Deleting these rows would have caused meaningful data loss, so missing dates were instead filled with Excel’s default placeholder date (year 1900) to keep every transaction record intact.
- This decision was later revisited during the Power BI phase, where the artificial 1900 dates were identified and excluded through a newly created Order Date column (see Phase 2).

Sales Category (Calculated Column)

- Created a Sales Category column based on the Total Spent value.
- Used the IF() function to classify each transaction into Low, Medium, or High spending categories.
`=IF(E2<=9,"Low",IF(E2<=20,"Medium","High"))`

Item Category (Calculated Column)

- Created an Item Category column that groups each item into Beverages, Food, or Unspecified based on the item name.
`=IF(OR(B2="Coffee",B2="Smoothie",B2="Tea",B2="Juice"),"Beverages",IF(OR(B2="Cake",B2="Cookie",B2="Salad",B2="Sandwich"),"Food","Unspecified"))`

Order Size (Calculated Column)

- Created an Order Size column that classifies each order as Single Item, Multiple Items, or Bulk Order based on Quantity.
`=IF(C2<=1,"Single Item",IF(C2<=3,"Multiple Items","Bulk Order"))`

Service Tax (Calculated Column)

- Created a Service Tax column calculated as 5% of Total Spent.
`=E2*0.05`

Discount Flag (Calculated Column)

- Created a Discount Flag column that assigns a discount percentage based on Order Size: 15% for Bulk Order, 10% for Multiple Items, and 0% for Single Item.
`=IF(K2="Bulk Order",0.15,IF(K2="Multiple Items",0.1,0))`
- Stored as a numeric percentage value throughout, not a Yes/No text flag.

Summary Statistics

- Built a Summary Statistics sheet to sanity-check the cleaned data before importing into Power BI.
- Total Revenue calculated using `=SUM(Café_Sales_Cleaned!E:E)`.
- Total Transactions calculated using `=COUNTA(Café_Sales_Cleaned!A:A)-1`.
- Average Order Value calculated using `=AVERAGE(Cafe_Sales_Cleaned!E:E)`.

Pivot Tables

- Created Pivot Tables to summarize and analyze the cleaned dataset.
- Generated a Quantity and Total Spent by Item Category Pivot Table to compare sales volume across Beverages and Food, with any remaining items grouped separately for reference.
- Created a Total Spent by Location × Payment Method Pivot Table to cross-tabulate revenue by store location and payment type.
- Developed a Total Spent by Year Pivot Table, which revealed that some transactions were grouped under the year 1900 instead of 2023 — confirming that placeholder dates used to fill missing Transaction Date values were present in the dataset.
- This finding directly informed the Order Date correction carried out during the Power BI phase.

Phase 2: Data Modeling and Visualization using Power BI

Data Import

- Imported the cleaned Excel dataset into Power BI.
- Verified the data for modeling and reporting.

Order Date Correction

- Addressed the 1900 placeholder date issue found in the Excel pivot analysis by creating a new column, Order Date, to hold only valid, real transaction dates.
- Excluded records affected by the artificial 1900 placeholder date from the reporting model, since using them directly would have skewed monthly trends, Year-to-Date calculations, and other date-based visuals.
- Retained the original Transaction Date column in the table but intentionally left it unused throughout the report — Order Date became the single source of truth for all date-based analysis.
- After this correction, the fact table used across the dashboard contains 9,540 clean transaction records, down from the original 10,000.

Data Normalization

- Created a Café Sales fact table containing all cleaned transaction records.
- Created a Dim Date calculated table using DAX to hold a single, unique list of order dates:
Dim Date = VALUES(Café Sales[Order Date])
- Dim Date contains 365 rows with Year, Month, Month Num, Quarter, Weekday, Weekday Num, Day of Month, Order Date, and an Order Date Hierarchy.
- Created a dedicated _Measures table to store and organize all DAX measures separately from the data tables.

Data Transformation

- Removed the default 00:00:00 time component from Order Date via Column Tools → Format for cleaner display in tables and axis labels.
- Fixed the Month sort order dashboard-wide using Sort by Column → Month Num.
- Unchecked "Unknown" and "Unspecified" across all slicers and applied filters, so every visual and chart on every report page reflects only identifiable categories.

Relationship Modeling

- Established a One-to-Many (1:*) relationship between Dim Date and Café Sales using the Order Date column.
- Applied Single Cross Filter Direction to optimize data filtering and reporting.

Executive Overview Dashboard Development

- Designed and developed a Cafe Sales Executive Overview page to provide a high-level overview of business performance.
- Created DAX measures to calculate key business metrics, including:
 1. Total Revenue
 2. Total Profit
 3. Total Transactions
 4. Average Selling Price
 5. Sales Growth
- Displayed the DAX measures using a single multi-value KPI card for a compact, consistent header.
- Developed interactive visualizations, including:
 1. Average Discount Rate (Gauge Chart)
 2. Revenue Share by Payment Method (Donut Chart)
 3. Revenue by Item Category (Bar Chart)
 4. Units Sold by Weekday and Category (Clustered Column Chart)
 5. Revenue vs. Previous Month Trend (Line Chart)
- Added interactive slicers for Month, Weekday, Payment Method, and Item Category to enable dynamic filtering.
- Designed a professional dashboard layout with a consistent pink/magenta theme, navigation arrows, cafe branding, and formatting to improve readability.

Customer Analysis Dashboard

- Created a new report page named Customer Analysis.
- Developed DAX measures: Total Transactions, Average Selling Price, Max Transaction Value, Min Transaction Value, and Total Units Sold.
- Stored all measures in the _Measures table for better organization.
- Added a single KPI card to display key customer metrics.
- Implemented interactive slicers for Location, Month, Payment Method, and Discount Flag.
- Created the following visualizations:
 1. Top Transactions by Value (Table)
 2. Spending Pattern by Transaction (Scatter Chart)
 3. Transactions by Order Size (Pie Chart)
 4. Revenue by Location (Bar Chart)
 5. Transaction Volume Trend (Area Chart)
- Maintained the same navigation panel and overall design as the Executive Overview page for consistency.

- Formatted KPI values by removing unnecessary decimal places and using K notation for improved readability.

Product Analysis Dashboard

- Created a Product Analysis page.
- Developed DAX measures for Total Units Sold, Average Unit Price, Total Revenue, Total Profit, and Item Revenue Rank.
- Added slicers for Quarter, Month, Sales Category, and Item Category.
- Designed visuals to analyze revenue build-up by category, unit sales by category and sales tier, revenue share by sales category, monthly item revenue trend, and average unit price trend:
 1. Revenue Build-up by Item Category (Waterfall Chart)
 2. Units Sold by Category and Sales Tier (100% Stacked Column Chart)
 3. Revenue Share by Sales Category (Pie Chart)
 4. Item Revenue Trend by Month (Stacked Column Chart)
 5. Average Unit Price Trend (Line Chart)
- Added a highlighted Top Revenue Item card, which identified Salad as the top-selling item by revenue.
- Used professional chart titles and formatted KPI values by removing unnecessary decimal places.
- Maintained consistent navigation buttons, layout, icons, and overall dashboard design.

Business Insight Dashboard

- Created a Business Insight page to provide a concise overview of key business performance and insights.
- Developed DAX measures for Total Revenue, Total Profit, Net Revenue Margin, Sales Growth, and Total Service Tax.
- Added a KPI card to display the key business metrics.
- Created the following visualizations:
 1. Revenue Breakdown by Category and Payment (Decomposition Tree)
 2. Revenue Share by Payment Method (Donut Chart)
 3. Revenue by Day of Month (Line Chart)
 4. Revenue vs. Previous Month Target (KPI/Target Visual)
- Added a Key Business Insights section and a Strategic Recommendations section to highlight important findings and next steps.
- Renamed the Net Revenue Margin measure from its original name, Profit Margin, since the figure it produces reflects only Service Tax being deducted from revenue, not the business's full cost structure.
- Applied professional chart titles, consistent formatting, and suitable number formatting.
- Maintained the Page Navigator, layout, icons, and overall dashboard design for consistency with the other pages.

Item Details (Hidden Drill through Page)

- Created a hidden drill through page, Item Details, accessible only by drilling through from the Product Analysis page.
- Displays a KPI card scoped to the selected item or category: Total Revenue, Total Profit, and Total Units Sold.
- Created supporting visuals: a bar chart (Total Revenue by Item Category), a table (Item Breakdown by Payment Method), and a line chart (Total Revenue by Month).
- Created a single shared Reset bookmark, applied across all report pages, to restore the default dashboard state after interacting with slicers and filters.

Findings & Insights

1. Overall Sales Performance

- The cafe generated approximately \$85.25K in total revenue across roughly 9,540 valid transactions.
- Approximately 29K units were sold, with an average selling price and average unit price of \$2.95.
- Revenue grew 9.26% compared to the previous month.

2. Customer Performance

- Transaction values ranged from a minimum of \$1.00 to a maximum of \$25.00 per transaction.
- By order size, Multiple Items transactions made up the largest share at 43.1%, followed by Bulk Orders at 38.66% and Single Item purchases at 18.24%.
- Revenue is split between In-store and Takeaway locations, with transaction volume trending gently upward across the year.

3. Product Performance

- Salad was identified as the top revenue-generating item.
- Food and Beverages both contribute meaningfully to overall revenue, with Food showing a modestly higher contribution.
- By sales tier, the Medium category accounts for the largest share of revenue at around 58%, followed by Low at around 35% and High at around 7%.
- Average unit price remained relatively stable across the year, staying close to \$2.95–\$3.00 per month.

4. Business Insights

- Payment method usage is nearly evenly balanced across Digital Wallet (33.12%), Credit Card (33.51%), and Cash (33.37%), indicating no single payment channel dominates.
- Total Service Tax collected was approximately \$4.3K, the key deduction separating Total Revenue from Total Profit.
- Net Revenue Margin sits at approximately 95%, reflecting only Service Tax being deducted — not a full operating margin, since it excludes ingredient, labor, and overhead costs.
- Revenue tracked slightly ahead of its previous-month target (\$7.22K achieved vs. a \$7.02K goal, +2.86%).

Overall Insight

The cafe demonstrates healthy, steadily growing sales performance, generating approximately \$85.25K in revenue with 9.26% month-over-month growth. Salad leads product revenue, while Food and Beverages contribute in reasonably close proportion to overall sales. Customer spending patterns lean toward multi-item and bulk orders rather than single-item purchases, and payment method usage remains evenly balanced across Cash, Credit Card, and Digital Wallet. Total Service Tax remains the main factor separating Total Revenue from Total Profit, and monthly performance continues to track slightly ahead of target.